

Manuscripts considered — aml-mds-related-rr-tp53-aberrant-hla-pe nding-x7q2-rerun2

Master inventory: every paper reviewed in this case

Generated 2026-08-21

Manuscripts considered — aml-mds-related-rr-tp53-aberrant-hla-p

Master inventory: 36 clinical (27 included, 9 considered & excluded) + 33 pre-clinical (29 included, 4 considered & excluded) + 38 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Krakow/Bleakley (2026) — Blood Advances

Reference: [PMID 42392183](#) · **Type:** clinical · **Inclusion:** excluded — Narrative review without extractable primary effect estimates; the TSC-100/TSC-101 (ALLOHA) results it summarises remain conference-abstract level, and that protocol excludes prior allo-HCT so this patient could not enrol regardless.

- **Intervention:** HA-1 / HA-2 minor-histocompatibility-antigen TCR-T (class review, incl. TSC-100/TSC-101)
- **Notes:** Excluded: Narrative review without extractable primary effect estimates; the TSC-100/TSC-101 (ALLOHA) results it summarises remain conference-abstract level, and that protocol excludes prior allo-HCT so this patient could not enrol regardless. — Read to check whether any peer-reviewed TSC-100/101 efficacy numbers exist. They do not; the review cites early-phase safety and persistence in general terms.

Lane/Pemmaraju (2026) — Blood

Reference: [PMID 42413008](#) · **Type:** clinical · **Inclusion:** excluded — BPDCN-only phase 2; the AML-relevant tagraxofusp combination evidence from the same NCT03113643 programme is already captured in Lane 2024 Blood Advances.

- **Intervention:** Tagraxofusp + azacitidine + venetoclax in BPDCN
- **Notes:** Excluded: BPDCN-only phase 2; the AML-relevant tagraxofusp combination evidence from the same NCT03113643 programme is already captured in Lane 2024 Blood Advances. — Read for the transplant-bridging signal (63% of the frontline cohort reached allo-HCT), which supports the sequencing logic but adds no AML effect estimate.

Pemmaraju/Daver (2026) — Journal of Clinical Oncology

Reference: [PMID 41671533](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pivkemab sunirine (CD123 ADC) in BPDCN — approval-supporting data
- **Indication / model:** Blastic plasmacytoid dendritic cell neoplasm, frontline and relapsed/refractory (any); 84 patients: 33 frontline (22 de novo, 20 forming the primary analysis population) and 51 relapsed/refractory. Median age 72.
- **Design:** Phase 1/2, open-label, multicentre (NCT03386513)
- **n:** 84
- **Effect size:** 75 % — Composite complete response rate in frontline de novo BPDCN
- **Variance:** 95% CI 51–91
- **Toxicities:** Peripheral oedema (n=84, 54%); Peripheral oedema G $\geq$ 3 (n=84, 12%); Fatigue (n=84, 26%); Infusion-related reaction (n=84, 26%); Neutropenia G $\geq$ 3 (n=84, 16%); Thrombocytopenia G $\geq$ 3 (n=84, 14%); Pneumonia G $\geq$ 3 (n=84, 6%)
- **Case fit:** cross_tumor_only

- **Notes:** The dataset behind the May 2026 FDA approval, which is in BPDCN and not AML. Relevant here only as evidence that the CD123 ADC is now commercially available and therefore reachable off-label or by expanded access.

Azevedo/Kadia (2026) — Leukemia & Lymphoma

Reference:PMID 41173347 · **Type:** clinical · **Inclusion:** excluded — Wrong population and line — frontline core-binding-factor AML in fit patients, which is the favourable-cytogenetics group where gemtuzumab benefit is concentrated. This patient has complex adverse karyotype and refractory disease.

- **Intervention:** CLAG-GO (cladribine, high-dose cytarabine, gemtuzumab ozogamicin) frontline in core-binding-factor AML
- **Notes:** Excluded: Wrong population and line — frontline core-binding-factor AML in fit patients, which is the favourable-cytogenetics group where gemtuzumab benefit is concentrated. This patient has complex adverse karyotype and refractory disease. — Read while looking for any published CLAG-GO efficacy. NCT04050280, the R/R CLAG-GO phase 2 that fits this case, is still recruiting with no results posted, so the regimen currently has no direct evidence in refractory AML.

Opydo/Kolaczowska (2026) — Biochem Biophys Res Commun

Reference:PMID 41980559 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin combined with BCL-2 and PI3K inhibition
- **Indication / model:** four AML cell lines of differing phenotype treated with gemtuzumab ozogamicin alone and with selective BCL-2 and/or PI3K inhibition
- **Design:** Calicheamicin-induced double-strand breaks trigger the intrinsic apoptotic pathway, which BCL-2 restrains; removing that brake lowers the damage threshold for commitment to apoptosis.
- **n:** 4 AML cell lines
- **Effect size:** moderate — Gemtuzumab ozogamicin caused G2/M arrest and apoptosis, but the magnitude varied widely across the four lines and did not track CD33 expression. Adding BCL-2 and/or PI3K inhibition increased the pro-apoptotic response.
- **Variance:** vs untreated cells; single-agent gemtuzumab ozogamicin; single-agent inhibitors; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Four cell lines, in vitro only, and its no-correlation-with-CD33 result sits against the 45-specimen correlation in Laszlo 2019 — the small line panel is the more likely explanation. The BCL-2 synergy it reports is less useful here than it looks, since this patient has already progressed through venetoclax.

Nadorp/Eisfeld (2026) — Blood Cancer Discov

Reference:PMID 42160138 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Myeloid sarcoma — site-specific clonal evolution and microenvironmental adaptation
- **Indication / model:** paired myeloid sarcoma and medullary AML patient specimens profiled by multiomic sequencing, with matched circulating tumour DNA
- **Design:** Extramedullary deposits arise from medullary AML but then evolve separately, acquiring RAS pathway activation and a transcriptional programme fitted to the soft-tissue microenvironment rather than the marrow niche.

- **Effect size:** Myeloid sarcoma carried universal molecular evolution away from the marrow clone, with RAS pathway activation as a recurring hallmark and transcriptional signatures reflecting extramedullary adaptation. Circulating tumour DNA captured the molecular composition of the extramedullary lesion, which offers a non-invasive route to profiling a thigh mass.
- **Variance:** vs matched marrow AML from the same patients; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive multiomics with no therapeutic arm, and antigen expression (CD33, CD123) was not the focus — so it establishes that marrow and thigh may diverge without saying by how much for the targets in play. Practical implication: biopsy or ctDNA-profile the thigh lesion rather than assuming the marrow panel represents it.

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-102-A01 / TSC-102-A03 (CD45 peptide-HLA-directed TCR-T)
- **Indication / model:** Haematologic malignancy after allogeneic HCT (planned first-in-human) (2L+); HLA-A01:01 or HLA-A03:01 restriction; CD45-derived peptide
- **Design:** IND cleared for two allele-matched candidates; phase 1 initiation guided for 2H 2026. No registry record posted at time of screen.
- **Effect size:** — — Not yet defined (pre-first-in-human)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BSB-2002 (mutant-NPM1-directed TCR-T)
- **Indication / model:** Relapsed or refractory NPM1-mutated AML, MRD-positive after prior remission (2L+); HLA-A*02:01 positive AND NPM1 mutation type A, D, G or H
- **Design:** Multicentre phase 1 dose finding of a TCR-modified T cell product given with standard of care
- **n:** 19
- **Effect size:** — — Safety and dose
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FH-WT1-E50 TCR-T + azacitidine
- **Indication / model:** MRD-positive AML in morphologic remission (2L+); HLA-A*02:01 confirmed by typing; WT1 expression by IHC on archival tissue
- **Design:** Phase 1 of autologous CD4+/CD8+ T cells transduced with a WT1-specific TCR plus a CD8ab

co-receptor, given with azacitidine

- **n:** 9
- **Effect size:** — — Safety and dose; MRD clearance
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pivekimab sunirine + venetoclax + azacitidine vs placebo + venetoclax + azacitidine
- **Indication / model:** Newly diagnosed AML in adults ineligible for intensive chemotherapy (1L); CD123-positive
- **Design:** Randomised phase 2/3 of pivekimab added to venetoclax plus azacitidine against placebo
- **n:** 660
- **Effect size:** — — Composite complete remission and overall survival
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** 131I-apamistamab conditioning + fludarabine/cyclophosphamide/TBI + allo-HCT
- **Indication / model:** Relapsed or refractory AML with active disease, unsuitable for myeloablative conditioning (2L+); CD45 expression by flow cytometry; 8/8 HLA-matched related or unrelated donor
- **Design:** Adaptive, operationally seamless phase II/III of 131I-apamistamab-led conditioning before allogeneic HCT
- **n:** 306
- **Effect size:** — — Durable complete remission and overall survival
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Mazziotta/Chapuis (2025) — Nature Communications

Reference: [PMID 40473616](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** WT1-TCRC4 CD8+ T cells (TTCR-C4, HLA-A*02:01-restricted), active disease post-HCT
- **Indication / model:** Relapsed and/or refractory AML after allogeneic HCT (2L+); 15 HLA-A2-positive patients with relapsed/refractory AML post-HCT enrolled 2013-2019. At infusion two had overt disease, five were MRD-positive, and the rest had no evaluable disease.
- **Design:** Phase 1/2, single-arm (NCT01640301)
- **n:** 15
- **Effect size:** 242 days — Overall survival
- **Toxicities:** Dose-limiting toxicity G_≥3 (0/15, 0%); Product-related serious AE G_≥3 (0/15, 0%); Acute GVHD G3 (1/15, 6.7%); Chronic GVHD (moderate) (1/15, 6.7%)

- **Case fit:** partial
- **Notes:** The negative counterpart to Chapuis 2019: in active post-HCT disease the same TCR did not improve outcomes over historical controls, with the transferred cells skewing to an NK-like terminally differentiated state. Safety detail from PMC12141728.

Thomas/Schendel (2025) — Cancers (Basel)

Reference:PMID 41008812 · **Type:** clinical · **Inclusion:** included

- **Intervention:** MDG1011 (PRAME-directed, HLA-A*02:01-restricted autologous TCR-T)
- **Indication / model:** Relapsed/refractory AML, MDS and multiple myeloma (2L+); 13 HLA-A*02:01-positive patients enrolled, 9 infused. Advanced disease stage and rapid progression in the R/R AML subset limited what could be read out.
- **Design:** First-in-human phase 1, single-arm (CD-TCR-001)
- **n:** 9
- **Effect size:** 22.2 % — Measurable clinical response
- **Toxicities:** Dose-limiting toxicity G_≥3 (0/9, 0%); CRS (2/9, 22.2%)
- **Case fit:** partial
- **Notes:** Medigene stopped the programme after phase 1. The authors' own reading is that tumour burden was the limiting factor, which matters for a patient at 85% marrow blasts.

Abedin/Atallah (2025) — Leukemia

Reference:PMID 39955432 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Lintuzumab-Ac225 (225Ac anti-CD33) after CLAG-M salvage
- **Indication / model:** Relapsed/refractory AML (2L+); 26 patients (21 in 3+3 escalation, 5 at the RP2D); 86.7% of evaluable patients had high-risk disease. Separate readouts for TP53-mutated and prior-venetoclax subgroups.
- **Design:** Phase 1 dose-escalation, single-arm
- **n:** 26
- **Effect size:** 56.6 % — Composite complete remission (CR/CRi)
- **Toxicities:** Febrile neutropenia G3-4 (n=26, 65.4%); White blood cell count decreased G3-4 (n=26, 50%)
- **Case fit:** strong
- **Notes:** The most directly transferable CD33 datapoint in this set: R/R AML, high-risk disease, and separate 50% TP53-mutated and 38.5% prior-venetoclax subgroup rates. Abstract-level safety only; no per-term grade table beyond the two events listed.

Schram/Dumbrava (2025) — Future Oncology

Reference:PMID 40932470 · **Type:** clinical · **Inclusion:** excluded — Trial-protocol paper with no efficacy results; PYNACLE enrolls solid tumours requiring RECIST-measurable disease, so a marrow-based leukaemia cannot contribute. Interim response rates circulating for this drug come from sponsor press releases rather than peer-reviewed publication.

- **Intervention:** Rezatapopt (PC14586, TP53 Y220C-selective p53 reactivator)
- **Notes:** Excluded: Trial-protocol paper with no efficacy results; PYNACLE enrolls solid tumours requiring RECIST-measurable disease, so a marrow-based leukaemia cannot contribute. Interim response rates circulating for this drug come from sponsor press releases rather than peer-reviewed publication. — Read to

establish whether any peer-reviewed rezatapopt effect size exists. None does. NCT06616636, the AML/MDS trial, has only just opened.

Carter/Andreeff (2025) — Blood

Reference: [PMID 40608889](#) · **Type:** clinical · **Inclusion:** excluded — Preclinical (cell lines, primary samples, xenografts); no patients treated.

- **Intervention:** Rezatapopt in TP53-Y220C AML — mechanism study
- **Notes:** Excluded: Preclinical (cell lines, primary samples, xenografts); no patients treated. — Kept in the audit trail because it tempers the Y220C hypothesis: rezatapopt restored wild-type p53 conformation in Y220C AML but induced limited or no apoptosis unless combined with venetoclax or MDM2/XPO1 inhibition.

Gyurkocza/Giralt (2025) — Journal of Clinical Oncology

Reference: [PMID 39298738](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** 131I-apamistamab (Iomab-B) targeted conditioning before allogeneic HCT
- **Indication / model:** Active relapsed/refractory AML, age 55 and over (2L+); 153 patients randomised (76 apamistamab, 77 conventional care). Active disease at randomisation was required. Prior allogeneic or autologous HSCT was an exclusion, so this patient could not have enrolled.
- **Design:** Randomised open-label phase 3 (NCT02665065)
- **n:** 153
- **Effect size:** 17.1 % — Durable complete remission lasting 180 days
- **Variance:** 95% CI 9.4–27.5; $p < 0.0001$
- **Toxicities:** Any treatment-related AE G $\geq$ 3 (n=76, 59.7%); Febrile neutropenia G $\geq$ 3 (n=76, 18.1%); Mucositis G $\geq$ 3 (n=66, 15.2%); Acute GVHD G1-2 (n=66, 27.3%); Acute GVHD G $\geq$ 3 (n=66, 9.1%); Sepsis G $\geq$ 3 (n=66, 6.1%); Non-relapse mortality (100 day) G5 (n=76, 12.2%); Treatment-related death G5 (3/72, 4.2%)
- **Case fit:** partial
- **Notes:** The randomised proof that a patient with active leukaemia can be transplanted when conditioning is antigen-targeted; the 0% control arm is what makes the comparison stark. Overall survival did not separate (HR 0.99), which crossover dilutes. Prior HCT was excluded, so this informs mechanism rather than eligibility. Safety from PMC11709001.

Bürdek/Schendel (2025) — Cancers (Basel)

Reference: [PMID 39858024](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MDG1011 (PRAME-directed, HLA-A*02:01-restricted autologous TCR-T)
- **Indication / model:** preclinical MDG1011 batches from enriched CD8+ T cells of healthy donors, plus 13 patient-derived investigational medicinal products; on-target and off-target panels of tumour and healthy-tissue test cells
- **Design:** A recombinant TCR against the PRAME100-108 VLD peptide on HLA-A*02:01 redirects autologous CD8+ T cells to PRAME-expressing myeloid blasts; PRAME is largely absent from normal adult tissue outside testis.
- **n:** 2 preclinical batches; 13 patient IMPs (manufacturing success 92%); 9 IMPs administered
- **Effect size:** moderate — Batches expressed the VLD-TCR at high density and produced antigen-specific IFN-gamma and cytotoxicity with no signal of on-target/off-tumour or off-target recognition. Manufacturing

succeeded in 12 of 13 heavily pretreated patients, which is the practical question for a patient at 85% marrow blasts.

- **Variance:** vs healthy-tissue test cell panels for on-target/off-tumour and off-target screening; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** No animal efficacy arm and no leukaemia-burden endpoint; specificity and manufacturability are the endpoints. PRAME expression on this patient's blasts has not been measured, and the target validator logged PRAME as not assessable.

Shi/Alatrash (2025) — Leukemia

Reference:PMID 39939820 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CG1/HLA-A2 x CD3 TCR-mimetic bispecific T-cell engager (preclinical basis for CBX-250)
- **Indication / model:** HLA-A2-positive primary AML blasts and AML cell lines in vitro, plus in-vivo AML models; normal bone marrow as the toxicity comparator
- **Design:** Cathepsin G is an azurophil-granule protease expressed more highly in AML blasts than in normal myeloid progenitors; its leader-sequence peptide CG1 is preferentially loaded onto HLA-A*02:01. A TCR-mimetic antibody against the CG1/HLA-A2 complex, formatted as a CD3 bispecific, recruits polyclonal T cells to that complex.
- **Effect size:** strong — The engager bound CG1/HLA-A2 monomers with high affinity and killed HLA-A2-positive primary AML and cell lines both in vitro and in vivo, with T-cell activation and cytokine release tracking tumour and drug exposure. No activity was seen against normal bone marrow.
- **Variance:** vs HLA-A2-negative leukaemia targets; normal bone marrow; T cells without engager; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Gated on HLA-A*02:01, which is unresolved here; CG1/HLA-A2 surface density on this patient's blasts is also unmeasured and has no routine assay. Crossbow Therapeutics co-authors, so read the safety margin with sponsor involvement in mind.

Yan/Alatrash (2025) — Leukemia

Reference:PMID 40437170 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CG1/HLA-A2 TCR-mimetic CAR T cells
- **Indication / model:** CG1/A2 TCR-mimic scFv-based CAR T cells against HLA-A2-positive AML and CML lines and primary samples, in vitro and in vivo; normal bone marrow haematopoiesis as safety readout
- **Design:** The same CG1/HLA-A2 TCR-mimic binder rebuilt as a CAR ectodomain, so recognition of a peptide-HLA complex drives CAR signalling rather than antibody-mediated T-cell recruitment.
- **Effect size:** strong — CG1/A2-CAR T cells killed HLA-A2-positive AML and CML in vitro and in vivo and left normal bone marrow haematopoiesis intact. The two formats together make the case that the CG1/A2 complex, not the format, is the discriminating element.
- **Variance:** vs untransduced T cells; HLA-A2-negative targets; normal bone marrow colony assays; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** A different modality from the CD3 engager on the trial list, so it corroborates the antigen rather than the

product. Same laboratory and same binder as Shi 2025, so the two are not independent replication.

Cole/Walter (2025) — Leukemia

Reference: [PMID 40119158](#) · Type: preclinical · Inclusion: included

- **Intervention:** Pivekimab sunirine — determinants of response (CD123 density, TP53, BCL-2 family)
- **Indication / model:** 13 human acute myeloid and lymphoblastic leukaemia cell lines, isogenic TP53-knockout/parental pairs, sublines overexpressing BCL-2, BCL-xL or MCL-1, and 9 primary AML samples
- **Design:** After CD123-mediated internalisation the IGN payload alkylates DNA, and the resulting death signal runs through a p53-dependent, BCL-2-family-gated apoptotic pathway — so both antigen density at the front end and apoptotic competence at the back end set the response.
- **n:** 13 cell lines; 5 isogenic TP53-KO/parental pairs; 9 primary AML samples
- **Effect size:** moderate — Cytotoxicity tracked CD123 expression ($R^2=0.4469$, $P=0.0125$), was significantly lower in TP53-knockout cells than in the matched wild-type parents across all 5 isogenic pairs, and was reduced in sublines overexpressing BCL-2 or BCL-xL. Three of the case's open variables — CD123 density, TP53 status, and prior venetoclax failure — are exactly the three modifiers this study identifies.
- **Variance:** vs parental TP53 wild-type lines; non-overexpressing parental sublines; unconjugated antibody and free payload; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Published as a short letter with no in-vivo arm, and the R^2 of 0.45 means antigen density explains under half the variance. TP53 knockout is a clean genetic model, not the mixed subclonal or copy-loss picture this patient probably has.

Carter/Andreeff (2025) — Blood

Reference: [PMID 40608889](#) · Type: preclinical · Inclusion: included

- **Intervention:** Rezatapopt (PC14586) in TP53-Y220C AML — mechanism and combination dependency
- **Indication / model:** TP53-Y220C AML cell lines and primary patient samples including stem/progenitor fractions; TP53-Y220C AML xenografts
- **Design:** Rezatapopt occupies the crevice created by the Y220C substitution and restores a wild-type p53 conformation, which reactivates p53 transcriptional targets. Two things then blunt the effect: MDM2 induced by the reactivated protein and XPO1-mediated nuclear export both damp transcription, and the refolded protein fails to bind BCL-2, BCL-xL and MCL-1 the way genuine wild-type p53 does — so the transcription-independent apoptotic arm is missing.
- **Effect size:** moderate — Rezatapopt converted p53-Y220C to a wild-type conformation and activated p53 targets, but induced limited or no apoptosis on its own in TP53-Y220C AML. Adding venetoclax supplied the missing BCL-2/BAX displacement, producing extensive death of blasts and stem/progenitor cells in vitro and prolonging xenograft survival; MDM2 or XPO1 inhibition restored the transcriptional arm.
- **Variance:** vs vehicle; rezatapopt alone versus rezatapopt plus venetoclax, MDM2 inhibition, or XPO1 inhibition; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Single-agent effect is weak — the strong readout needs venetoclax, and this patient has already progressed through venetoclax with FLAG-IDA, so the one combination that rescues rezatapopt in these models is the one she has already failed. A confirmed Y220C therefore opens a partial lever, not a clean one; PMV

Puzio-Kuter/Poyurovsky (2025) — Cancer Discov

Reference: [PMID 39945593](#) · Type: preclinical · Inclusion: included

- **Intervention:** Rezatapopt — discovery and characterisation of Y220C reactivation
- **Indication / model:** biochemical and structural analyses of p53-Y220C, in-vitro and in-vivo transcriptomics, and solid-tumour xenograft models as single agent and with immunotherapy
- **Design:** A designed small molecule binds the Y220C-specific surface pocket, corrects protein conformation, and restores sequence-specific DNA binding so the wild-type transcriptional programme reassembles.
- **Effect size:** strong — Conformational correction translated into recovery of multiple arms of the wild-type p53 programme and antitumour activity in preclinical models, alone and combined with immunotherapy. This is the paper that establishes the molecule; it does not test myeloid disease.
- **Variance:** vs vehicle; p53 wild-type and non-Y220C mutant comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Solid-tumour models throughout, with no AML or MDS system — the leukaemia-specific result, and the apoptotic shortfall it uncovers, comes from Carter 2025 instead. Written by PMV Pharmaceuticals employees.

Kruer/Padron (2025) — Blood

Reference: [PMID 40737597](#) · Type: preclinical · Inclusion: included

- **Intervention:** XPO1-mediated resistance to eprenetapopt plus azacitidine
- **Indication / model:** sequential genotyped samples from two phase 2 trials, a genome-wide CRISPR screen in TP53-mutated leukaemia cells, and a patient-derived xenograft of TP53-mutant MDS
- **Design:** XPO1 is overexpressed after eprenetapopt/azacitidine exposure and exports reactivated p53 out of the nucleus, which is necessary and sufficient for resistance; blocking nuclear export keeps p53 where it can act.
- **Effect size:** moderate — A genome-wide CRISPR screen nominated XPO1, and the trial samples confirmed XPO1 overexpression after treatment. Combining eprenetapopt with XPO1 inhibitors overcame resistance across model systems including a TP53-mutant MDS PDX.
- **Variance:** vs pretreatment paired samples; eprenetapopt alone; XPO1 inhibitor alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Relapse was near-universal in these trials unless transplant followed — the honest read is that p53 reactivation buys a window for a graft rather than durable control. XPO1 convergence with the Carter 2025 rezatapopt data is worth noting; the same export brake limits both compounds.

—/— (2025)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** BSB-1001 (HA-1 minor-histocompatibility-antigen TCR-modified T cells)
- **Indication / model:** Relapsed AML, relapsed ALL or MDS undergoing HLA-matched allogeneic HCT (2L+); HLA-A*02:01 AND HA-1-positive (H/H or H/R); 10/10-matched HA-1-negative (R/R) donor
- **Design:** Multicentre ascending-dose phase 1/2a of a donor-derived HA-1 TCR-T given with standard transplant
- **n:** 38

- **Effect size:** — — Treatment-emergent adverse events including GVHD and DLTs; cellular kinetics
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CBX-250 (TCR-mimetic bispecific T-cell engager against CG1/HLA-A*02:01)
- **Indication / model:** Relapsed or refractory AML, high-risk MDS, CMML or CML (CROSSCHECK-001) (2L+); Historical documented HLA-A02:01 *positivity*; *CG1 (cathepsin G-derived peptide) presented on HLA-A02:01*
- **Design:** Open-label phase 1 dose escalation, subcutaneous step-up dosing in 28-day cycles
- **n:** 72
- **Effect size:** — — Dose-limiting toxicity, RP2D/MTD, CRS frequency and severity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MT-401-OTS (off-the-shelf multiTAA-specific T cells)
- **Indication / model:** Relapsed AML or MDS (RAPID) (2L+); Off-the-shelf allogeneic product requiring $\geq 2/8$ HLA match; multi-tumour-antigen specificity
- **Design:** Phase 1 of banked allogeneic multi-antigen-specific T-cell products selected by partial HLA match
- **n:** 40
- **Effect size:** — — Safety
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tagraxofusp-erzs + azacitidine (post-HCT maintenance)
- **Indication / model:** CD123-positive AML or MDS, maintenance after first or second allogeneic HCT (2L+); CD123 positive by flow ($>20\%$ of the abnormal population above internal negative control) at any point before HCT
- **Design:** Phase 1 of post-transplant maintenance with tagraxofusp plus azacitidine, starting after day-30 marrow assessment
- **n:** 43
- **Effect size:** — — Safety and tolerability of maintenance; relapse
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tagraxofusp + cladribine + cytarabine
- **Indication / model:** CD123-positive relapsed or refractory AML (2L+); CD123 by flow or IHC with no minimum threshold for positivity
- **Design:** Phase 1/2 of tagraxofusp added to cladribine and cytarabine after venetoclax/HMA failure
- **n:** 20
- **Effect size:** — — Safety and response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CLL-1 / CD33 / CD123-directed CAR-T
- **Indication / model:** Acute myeloid leukemia expressing CLL-1, CD33 and/or CD123 (2L+); CLL-1, CD123 and/or CD33 confirmed by IHC or flow cytometry
- **Design:** Multi-centre phase I/II of single or combined CAR-T products directed at CLL-1, CD33 and CD123
- **n:** 10
- **Effect size:** — — Safety and response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) + azacitidine
- **Indication / model:** TP53 Y220C-mutant AML or MDS, relapsed or primary refractory (2L+); TP53 Y220C confirmed by CLIA-approved local testing at variant allele frequency >2%
- **Design:** Phase Ib of the Y220C-selective p53 reactivator rezatapopt combined with azacitidine
- **n:** 24
- **Effect size:** — — Safety, dose, response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Decitabine + FLAG-Ida followed by RIC allo-HCT (DEC-FLAG-Ida/RIC)
- **Indication / model:** Myeloid malignancy at high risk of relapse, including primary refractory AML and AML in

untreated or unsuccessfully treated relapse (2L+); None required beyond diagnosis; donor-specific antibody MFI <5000 for haploidentical donors

- **Design:** Phase 1/2 of sequential decitabine with FLAG-Ida followed immediately by reduced-intensity conditioning and allogeneic HCT
- **n:** 36
- **Effect size:** — — Safety and remission/transplant outcomes
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Krakow/Bleakley (2024) — Blood

Reference: [PMID 38683966](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** HA-1 TCR-T (donor-derived CD8+/CD4+ memory T cells, Bleakley/HighPassBio platform)
- **Indication / model:** Persistent or recurrent leukemia / myeloid neoplasm after allogeneic HCT (2L+); 9 allo-HCT recipients with disease recurrence after transplant, recipient HA-1(H)-positive and HLA-A*02:01-positive with an HA-1-disparate donor. Prior allo-HCT is an entry requirement rather than an exclusion.
- **Design:** Phase 1, single-arm, dose-escalation (NCT03326921)
- **n:** 9
- **Effect size:** 44.4 % — Complete remission achieved or maintained after lymphodepletion + HA-1 TCR-T
- **Toxicities:** CRS (0/9, 0%); ICANS (0/9, 0%); Dose-limiting toxicity G_≥3 (0/9, 0%); Acute GVHD (mild) G1-2 (2/9, 22.2%); Neutropenia G_≥3 (n=9); Infection G_≥3 (n=9); Infusion reaction G4 (1/9, 11.1%)
- **Case fit:** strong
- **Notes:** The only HA-1 programme whose protocol requires prior allo-HCT, which is why it survives the eligibility screen where TSC-100/101 and BSB-1001 do not. Safety data read from the PMC full text (PMC11406181); the paper reports events narratively rather than in a per-term grade table.

Lane/Pemmaraju (2024) — Blood Advances

Reference: [PMID 38052038](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tagraxofusp + azacitidine +/- venetoclax
- **Indication / model:** CD123-positive AML and high-risk MDS (1L); 56 adults enrolled; the untreated-AML triplet expansion was 26 patients, median age 71, all ELN adverse-risk, half TP53-mutated. 13 of the 26 carried a TP53 mutation.
- **Design:** Phase 1b, dose-finding with expansion (NCT03113643)
- **n:** 26
- **Effect size:** 69 % — Overall response (CR + CRi + MLFS) in untreated AML
- **Toxicities:** Capillary leak syndrome (7/37, 18.9%); Capillary leak syndrome (9/19, 47%); Thrombocytopenia G_≥3 (n=37, 45.9%); White blood cell count decreased G_≥3 (n=37, 45.9%); Neutrophil count decreased G_≥3 (n=37, 37.8%); Anaemia G_≥3 (n=37, 27%); Febrile neutropenia G_≥3 (n=37, 24.3%); Infection G_≥3 (n=37, 16.2%); Tumour lysis syndrome G_≥3 (n=37, 10.8%); 30-day all-cause mortality G5 (4/37, 10.8%)
- **Case fit:** partial
- **Notes:** Closest CD123 evidence to this case because half the cohort was TP53-mutated and all were ELN adverse-risk, but the population is untreated AML rather than post-transplant refractory disease. Safety extracted from PMC10837492.

Daver/Kantarjian (2024) — Lancet Oncology

Reference: [PMID 38423051](#) · Type: clinical · Inclusion: included

- **Intervention:** Pivekimab sunirine (IMGN632, CD123-directed antibody-drug conjugate)
- **Indication / model:** Relapsed or refractory CD123-positive AML (2L+); 91 enrolled overall (68 on the every-3-week schedule A); the efficacy estimate below is the 29 patients treated at the recommended phase 2 dose. ECOG 0-1 required.
- **Design:** First-in-human phase 1/2, dose escalation and expansion (NCT03386513)
- **n:** 29
- **Effect size:** 21 % — Overall response rate at the recommended phase 2 dose
- **Variance:** 95% CI 8–40
- **Toxicities:** Febrile neutropenia G $\geq$ 3 (3/29, 10%); Infusion-related reaction G $\geq$ 3 (2/29, 7%); Anaemia G $\geq$ 3 (2/29, 7%); Veno-occlusive disease G $\geq$ 3 (2/68, 2.9%); Treatment-related death G5 (1/68, 1.5%)
- **Case fit:** strong
- **Notes:** The single-agent activity estimate the board will want when weighing a CD123 ADC against cellular options. Veno-occlusive disease in a patient heading toward a second transplant is the specific risk to weigh.

Ma/Wang (2024) — Cancer Immunol Immunother

Reference: [PMID 39535595](#) · Type: preclinical · Inclusion: included

- **Intervention:** Loop33 x 123 dual CD33/CD123 CAR T cells
- **Indication / model:** loop-configuration bivalent CAR T cells from healthy-donor T cells; AML cell lines and primary AML cells in coculture; cell-derived xenografts using wild-type MOLM-13 and a mixture of MOLM-13-KO33 and MOLM-13-KO123 as an antigen-escape model
- **Design:** A single receptor carrying binders for both CD33 and CD123 fires when either antigen is present, so a blast that has lost one antigen is still recognised through the other.
- **Effect size:** moderate — Loop33 x 123 CAR T cells cleared AML lines and primary AML cells with higher degranulation and cytokine output than the monovalent constructs, and controlled the mixed CD33-knockout/CD123-knockout xenograft that single-antigen CARs could not. Loop orientation mattered: the 33 x 123 arrangement outperformed 123 x 33.
- **Variance:** vs monovalent CD33 and CD123 CAR T cells; untransduced T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The escape model is engineered knockout rather than the naturally heterogeneous antigen expression expected between this patient's marrow and her thigh sarcoma. No clinical product with this design is on the trial list; the relevant trial is a CLL-1/CD33/CD123 CAR (NCT04010877) with a different architecture.

—/— (2024)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** MB-dNPM1-TCR.1 (mutant-NPM1 TCR-T)
- **Indication / model:** Relapsed or refractory NPM1-mutated AML without standard options (2L+); HLA-A*02:01 positive AND NPM1 mutation recognised by dNPM1-TCR.1 with a monitoring qPCR available
- **Design:** Phase I/II dose finding of an autologous mutant-NPM1-directed TCR-T
- **n:** 29

- **Effect size:** — — Safety; first efficacy signal
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Allo-RevCAR01-T + R-TM123 (CD123 target module)
- **Indication / model:** Relapsed or refractory CD123-positive AML (2L+); CD123 expressed on $\geq 20\%$ of leukaemic cells at any point; HLA match at the B and C loci for the allogeneic platform
- **Design:** Phase 1a/1b of an allogeneic switchable reverse-CAR T cell platform dosed with a soluble CD123 target module
- **n:** 80
- **Effect size:** — — Safety, tolerability, dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BMS-986497 (ORM-6151, CD33-directed antibody-conjugated GSPT1 degrader)
- **Indication / model:** Relapsed or refractory AML or MDS, primary or secondary (2L+); Detectable CD33 expression
- **Design:** First-in-human phase 1 as monotherapy, with azacitidine, and with azacitidine plus venetoclax
- **n:** 105
- **Effect size:** — — Safety, dose-limiting toxicity, recommended dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GTB-3650 (anti-CD16 / IL-15 / anti-CD33 TriKE)
- **Indication / model:** High-risk or treatment-related MDS and relapsed/refractory AML (2L+); CD33 (target of the engager); NK-cell count $\geq 25/\mu\text{L}$ or ALC $\geq 200/\mu\text{L}$
- **Design:** Phase 1 dose finding of a tri-specific killer engager linking CD16 and CD33 with an IL-15 moiety
- **n:** 45
- **Effect size:** — — Safety and maximum tolerated dose
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Interferon gamma-1b + donor leukocyte infusion (DLI)
- **Indication / model:** AML or MDS relapsed after allogeneic transplant with measurable residual disease (2L+); 5-30% myeloblasts by morphology, or LAIP-positive blasts up to 30% by flow, or recurrent cytogenetic abnormality; DLI available from the original donor
- **Design:** Phase 2 of interferon-gamma-1b combined with donor leukocyte infusion in two cohorts by donor match
- **n:** 57
- **Effect size:** — — Response to DLI
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Ravandi/Mims (2023) — Blood Advances

Reference: [PMID 37647601](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Vibecotamab (XmAb14045, CD123 x CD3 bispecific)
- **Indication / model:** Relapsed/refractory AML (2L+); 120 patients dosed, 111 AML patients evaluable for response; 87 evaluable at the optimised dose of at least 0.75 microg/kg.
- **Design:** Phase 1 dose-escalation/expansion, single-arm
- **n:** 120
- **Effect size:** 9.0 % — Overall response rate (CR/CRi/MLFS/PR)
- **Toxicities:** CRS (102/120, 85%); CRS G $\geq$ 3 (32/120, 26.7%); Any treatment-emergent AE G $\geq$ 3 (105/120, 87.5%); Anaemia G $\geq$ 3 (31/120, 25.8%); Febrile neutropenia G $\geq$ 3 (30/120, 25%); Pneumonia G $\geq$ 3 (21/120, 17.5%); Treatment-related death G5 (1/120, 0.8%)
- **Case fit:** partial
- **Notes:** The second CD123 x CD3 readout and a considerably weaker one than flotetuzumab: 9% response with 27% grade 3+ CRS. Safety from PMC10632668.

Garcia-Manero/Sallman (2023) — Lancet Haematology

Reference: [PMID 36990622](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246) + venetoclax + azacitidine
- **Indication / model:** TP53-mutated AML, treatment-naïve (1L); 49 enrolled across cohorts; the response estimate is the 39 patients on the triplet. At least one pathogenic TP53 mutation was required, ECOG 0-2, median age 67.
- **Design:** Phase 1 dose-finding with cohort expansion (NCT04214860)
- **n:** 39
- **Effect size:** 38 % — Complete response with the triplet
- **Variance:** 95% CI 23–55
- **Toxicities:** Febrile neutropenia G $\geq$ 3 (23/49, 47%); Thrombocytopenia G $\geq$ 3 (18/49, 37%); Leukopenia G $\geq$ 3 (12/49, 25%); Anaemia G $\geq$ 3 (11/49, 22%); Treatment-related serious AE G $\geq$ 3 (13/49, 27%); Treatment-related death (sepsis) G5 (1/49, 2%)
- **Case fit:** weak

- **Notes:** Treatment-naïve AML, not relapsed disease, and the parallel randomised phase 3 of eprenetapopt plus azacitidine in TP53-mutant MDS missed its endpoint, after which the programme stopped. The 38% belongs to a population and a development path that no longer apply.

Issa/Stein (2023) — Nature

Reference:PMID 36922593 · **Type:** clinical · **Inclusion:** excluded — Biomarker mismatch and out of mechanism scope — menin inhibitors act on KMT2A-rearranged or NPM1-mutant leukemia. This patient has KMT2A amplification (4 copies) with no rearrangement and no reported NPM1 mutation, so the drug does not target any nominated feature.

- **Intervention:** Revumenib (SNDX-5613, menin inhibitor)
- **Notes:** Excluded: Biomarker mismatch and out of mechanism scope — menin inhibitors act on KMT2A-rearranged or NPM1-mutant leukemia. This patient has KMT2A amplification (4 copies) with no rearrangement and no reported NPM1 mutation, so the drug does not target any nominated feature. — Read specifically to confirm the eligibility biology before dropping the class. The dependency reported is on the menin-KMT2A protein interaction in rearranged disease, not on KMT2A copy number.

Pilunov/Efimov (2023) — Cancers (Basel)

Reference:PMID 36900382 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HA-1-specific transgenic TCR CD8+ T cells (independent TCR panel)
- **Indication / model:** CD8+ T cells with endogenous TCR knocked out and one of 13 HA-1-specific transgenic TCRs; cytotoxicity against primary leukaemia cells from HA-1+ patients
- **Design:** Transgenic HA-1/HLA-A*02:01 TCRs redirect CD8+ T cells against a haematopoiesis-restricted polymorphic peptide; knocking out the endogenous TCR removes mispairing and raises transgenic TCR surface density.
- **n:** 15 HA-1-positive patient samples (AML, T-ALL, B-ALL); 10 HA-1-negative or HLA-A*02-negative donor controls; 28-allele HLA panel for cross-reactivity
- **Effect size:** moderate — Lysis occurred against haematopoietic cells from HA-1-positive patients including AML, and was absent against HA-1-negative or HLA-A*02-negative donors. No cross-reactivity appeared across a 28-allele HLA panel.
- **Variance:** vs cells from HA-1-negative or HLA-A*02:01-negative donors; peripheral blood mononuclear cells across 28 common HLA alleles; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In vitro only, with reporter-line affinity ranking rather than in-vivo efficacy. Independent replication of the HA-1 target rather than of the specific TSC-100 or HighPassBio product.

Yang/Lu (2023) — Blood

Reference:PMID 36322939 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** C1Q+ macrophage-like leukaemia subset driving extramedullary infiltration
- **Indication / model:** single-cell RNA sequencing of paired bone marrow and extramedullary lesions, verified across multiple AML patients; patient-derived and cell-line-derived xenografts forming cutaneous and gastrointestinal nodules

- **Design:** MAFB-driven C1Q expression gives blasts tissue-infiltration capacity; C1Q engages the globular C1Q receptor on fibroblasts, which release TGF-beta1 and both attract the blasts and protect them under chemotherapy stress.
- **Effect size:** moderate — A C1Q+ macrophage-like subset was enriched in extramedullary lesions and detectable in marrow before extramedullary disease became clinically apparent, and it established cutaneous and gastrointestinal nodules in xenografts. Fibroblast crosstalk kept these cells alive under chemotherapy, which is a stromal explanation for sanctuary-site behaviour.
- **Variance:** vs C1Q-low leukaemia populations; marrow-only disease; xenografts without MAFB-driven C1Q expression; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Anchored on one index patient with leukaemia cutis before broader verification, and skin/gut rather than deep soft tissue. No C1Q-directed agent exists, so the value is explanatory: an extramedullary deposit can be protected by stroma independently of the drug's target.

Farge/Carrière (2023) — Cancer Res

Reference: PMID 37327406 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CD36-driven extramedullary dissemination and post-chemotherapy relapse
- **Indication / model:** cohort of 1,273 AML patients for the expression-outcome association, plus xenograft mouse models with CD36 inhibition after chemotherapy
- **Design:** CD36 acts here not as a lipid transporter but as a thrombospondin-1 receptor driving blast migration; CD36+ blasts adopt a senescence-like state that survives chemotherapy while retaining the ability to move into tissue.
- **n:** n=1,273 patients (association); xenograft arms for the intervention
- **Effect size:** moderate — CD36 expression associated with extramedullary dissemination, higher relapse risk after intensive chemotherapy, and shorter event-free and overall survival. CD36+ blasts were enriched after chemotherapy, and CD36 inhibition reduced dissemination and prolonged survival in chemotherapy-treated xenografts.
- **Variance:** vs CD36-low blasts; untreated and chemotherapy-only xenograft arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Post-FLAG-IDA enrichment of a migratory, chemo-surviving population is a plausible account of how a thigh sarcoma appeared in this patient, but CD36 was not measured on her blasts and no CD36-directed agent is clinically available.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** VIP943 (CD123-directed antibody-drug conjugate)
- **Indication / model:** Advanced CD123-positive AML, B-ALL or high-risk MDS after exhausting standard therapy (2L+); CD123 expression from a local laboratory; $\geq 5\%$ marrow or blood blasts
- **Design:** Open-label multicentre phase 1 dose escalation of a CD123 ADC, weekly and twice-weekly schedules
- **n:** 36
- **Effect size:** — — Safety, tolerability, preliminary antitumour activity
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Anti-CD33 CAR-T
- **Indication / model:** CD33-positive relapsed or refractory AML, de novo or secondary (2L+); CD33 positivity confirmed by flow cytometry or IHC within 90 days
- **Design:** Phase 1 of lentivirally transduced autologous CD33-specific CAR-T after lymphodepletion
- **n:** 27
- **Effect size:** — — Safety and dose-limiting toxicity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Venetoclax + FLAMSA (fludarabine/amsacrine/cytarabine) + treosulfan conditioning + allo-HCT
- **Indication / model:** MDS, CMML or secondary AML proceeding to allogeneic transplant (FLAMSAClax) (2L+); High-risk genetics including TP53, complex or adverse karyotype, IPSS-R/IPSS-M high or very high
- **Design:** Phase I/II adding venetoclax to sequential FLAMSA plus treosulfan conditioning before allogeneic transplant
- **n:** 38
- **Effect size:** — — Safety and efficacy of the sequential conditioning
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Mishra/Fernandez (2022) — Journal of Clinical Oncology

Reference: [PMID 35816664](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprex + azacitidine as post-transplant maintenance
- **Indication / model:** TP53-mutated AML and MDS after allogeneic HCT (maintenance); 33 of 55 transplanted patients started maintenance (14 AML, 19 MDS), median age 65. Screening began before transplant, so the cohort is enriched for patients who reached and tolerated HCT.
- **Design:** Phase 2, open-label, multicentre, single-arm (NCT03931291)
- **n:** 33
- **Effect size:** 12.5 months — Relapse-free survival
- **Toxicities:** Acute GVHD (4/33, 12%); Chronic GVHD (11/33, 33%); 60-day mortality G5 (2/33, 6%); 30-day mortality G5 (0/33, 0%)
- **Case fit:** partial
- **Notes:** The one eprex setting that maps onto this case's shape — TP53-mutant myeloid disease consolidated after allo-HCT — but it is single-arm with heavy selection: only 33 of 84 screened reached

maintenance.

Gambacorta/Vago (2022) — Cancer Discov

Reference: [PMID 35255120](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRC2 inhibition to restore HLA class II and rebuild donor T-cell recognition
- **Indication / model:** paired diagnosis/relapse primary AML samples and the corresponding patient-derived xenografts, profiled for gene expression, DNA methylation and chromatin accessibility; pharmacological PRC2 inhibition in vitro and in vivo
- **Design:** Loss of HLA class II at post-transplant relapse is enforced by PRC2-dependent closure of chromatin at the class II locus rather than by mutation, so an EZH2/PRC2 inhibitor can reopen it and restore surface expression.
- **Effect size:** strong — Class II loss tracked with PRC2-dependent reductions in chromatin accessibility, and PRC2 inhibitors restored HLA class II expression in relapsed AML both in vitro and in PDX models, with recovery of CD4+ T-cell recognition. Up to 40% of post-transplant relapses in this series were driven by nongenomic class II loss.
- **Variance:** vs vehicle-treated PDX and cultures; diagnosis-timepoint samples with intact HLA class II; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** No approved myeloid PRC2 inhibitor and no such trial on this case's list, so the actionable read is diagnostic: measure HLA class II on the blasts before committing to a DLI- or graft-dependent strategy. Restoring recognition also restores graft-versus-host risk.

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-100 (donor-derived HA-1-specific TCR-T)
- **Indication / model:** AML, MDS or ALL undergoing allogeneic PBSC transplant (ALLOHA / ALLOHA-2) (2L+); HLA-A02:01 positive; recipient HA-1-positive (HA1+/- or +/+); donor HA-1-negative or HLA-A02-negative
- **Design:** Multi-arm, non-randomised, concurrent-controlled umbrella; TCR-T given as consolidation after haploidentical, MMUD or MUD RIC transplant. Record now spans the pivotal ALLOHA-2 portion.
- **n:** 310
- **Effect size:** — — Dose-limiting toxicity and adverse events; disease-free survival vs concurrent control
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-101 (donor-derived HA-2-specific TCR-T)
- **Indication / model:** AML, MDS or ALL undergoing allogeneic PBSC transplant (ALLOHA / ALLOHA-2) (2L+); HLA-A02:01 positive; recipient HA-2-positive; donor negative for all HLA-A02 alleles; HLA-A*02:07 excluded
- **Design:** Sister arm of the same umbrella for HA-1-negative / HA-2-positive recipients; TCR-T consolidation after

RIC transplant

- **n:** 310
- **Effect size:** — — Dose-limiting toxicity and adverse events; disease-free survival vs concurrent control
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Vibecotamab (XmAb14045, CD123xCD3 bispecific)
- **Indication / model:** MRD-positive AML or MDS after hypomethylating-agent failure (2L+); CD123 expression on residual leukaemia at $\geq 20\%$
- **Design:** Single-arm phase 2 of a CD123xCD3 bispecific antibody with steroid premedication
- **n:** 51
- **Effect size:** — — MRD conversion / response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Lulla/Leen (2021) — Blood

Reference: [PMID 33270816](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Donor-derived multi-leukemia-antigen-specific T cells (mLSTs targeting PRAME, WT1, survivin, NY-ESO-1)
- **Indication / model:** AML/MDS after allogeneic HCT, high relapse risk or overt relapse (2L+); 25 patients infused, 17 at high risk of relapse and 8 with relapsed disease. Products were generated from 29 HCT donors and were not HLA-restricted to a single allele.
- **Design:** Phase 1, dose-escalation, single-arm (NCT02494167)
- **n:** 25
- **Effect size:** 25 % — Objective response in the active-disease cohort
- **Toxicities:** Acute GVHD $G \geq 3$ (0/25, 0%); Chronic GVHD (extensive) (0/25, 0%)
- **Case fit:** partial
- **Notes:** This is the ADSPAM protocol whose Group B admits post-transplant relapse and counts extramedullary sites as qualifying disease, which is why the thigh sarcoma does not disqualify. Response numbers in the active-disease arm are 2 of 8.

Uy/DiPersio (2021) — Blood

Reference: [PMID 32929488](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Flotetuzumab (MGD006, CD123 x CD3 DART)
- **Indication / model:** Relapsed/refractory AML, primary induction failure or early relapse (2L+); 88 adults with R/R AML overall (42 dose-finding, 46 at the recommended phase 2 dose). The efficacy estimate is the 30 primary-induction-failure or early-relapse patients treated at the recommended dose.
- **Design:** Multicentre open-label phase 1/2 (NCT02152956)

- **n:** 30
- **Effect size:** 26.7 % — CR / CR with partial haematological recovery in primary induction failure or early relapse
- **Toxicities:** IRR/CRS (48/50, 96%); IRR/CRS G $\geq$ 3 (4/50, 8%); Anaemia G $\geq$ 3 (n=88, 28.4%); Thrombocytopenia G $\geq$ 3 (n=88, 20.5%); Lymphopenia G $\geq$ 3 (n=88, 18.2%); Hypophosphataemia G $\geq$ 3 (n=88, 14.8%); Hypokalaemia G $\geq$ 3 (n=88, 13.6%); Treatment-related death G5 (0/88, 0%)
- **Case fit:** partial
- **Notes:** MacroGenics terminated the programme for business reasons in 2022, so the drug is not obtainable. It stays in the dossier because 26.7% CR/CRh in exactly the chemorefractory population this patient occupies is the best CD123 x CD3 benchmark on record. Safety from PMC7885824.

Qin/Fry (2021) — Journal for ImmunoTherapy of Cancer

Reference:PMID 34531250 · **Type:** clinical · **Inclusion:** excluded — Preclinical only — in vitro and xenograft construct comparison, no patients treated.

- **Intervention:** CD33 CAR construct optimisation (lintuzumab-CD28/CD3zeta)
- **Notes:** Excluded: Preclinical only — in vitro and xenograft construct comparison, no patients treated. — Read to see whether any clinical CD33 CAR-T efficacy exists beyond the 2015 single-patient report. It does not; this paper is the preclinical basis for the current phase 1.

Qin/Fry (2021) — J Immunother Cancer

Reference:PMID 34531250 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CD33-directed CAR T cells — construct optimisation (lintuzumab-CD28/CD3zeta)
- **Indication / model:** six CD33 CAR constructs (three scFvs x CD28 or 4-1BB costimulation) in lentivirally transduced human T cells; CD33+ AML cell lines and childhood AML patient-derived xenografts
- **Design:** The costimulatory endodomain sets CAR T expansion and persistence, and the scFv sets both epitope and tonic signalling — so myeloid-antigen CAR potency and toxicity are construct-dependent rather than antigen-dependent.
- **n:** 6 CAR constructs; AML cell line and PDX in-vivo models
- **Effect size:** moderate — CD28-based constructs suppressed leukaemia more consistently than 4-1BB-based constructs in both cell-line and PDX models, and lintuzumab-CD28/CD3zeta was carried forward for clinical translation. Gemtuzumab-scFv-based constructs were unexpectedly toxic in vivo despite in-vitro activity, a mismatch worth remembering when in-vitro data are the only data available.
- **Variance:** vs untransduced T cells; head-to-head comparison across scFv and costimulatory domain; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PDX models are paediatric AML, not post-transplant adult AML-MR, and CD33 CAR T cells carry unresolved myeloablation risk in a patient whose only route to durable control is another graft. Carries a 2021 publisher erratum, not a retraction.

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NTLA-5001 (WT1-directed TCR-T)

- **Indication / model:** AML with detectable disease after first-line therapy (2L+); HLA-A*02:01 positive; WT1
- **Design:** Single-dose phase 1/2a of a CRISPR-engineered autologous WT1-directed TCR-T; terminated after 6 patients
- **n:** 6
- **Effect size:** — — Safety and tolerability
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Vadakekolathu/Rutella (2020) — Blood Advances

Reference: [PMID 33057635](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Flotetuzumab in TP53-abnormal relapsed/refractory AML
- **Indication / model:** Relapsed/refractory AML with TP53 mutation and/or 17p deletion (2L+); 15 of 45 flotetuzumab-treated patients with marrow transcriptomic profiling carried TP53 mutations and/or 17p deletion. Responders had higher baseline tumour-inflammation, FOXP3, CD8 and PD1 signature scores.
- **Design:** Correlative analysis nested in the phase 1/2 trial (NCT02152956)
- **n:** 15
- **Effect size:** 47 % — Complete response (<5% marrow blasts) in TP53-abnormal R/R AML
- **Case fit:** partial
- **Notes:** Post-hoc biomarker analysis on 15 patients, so the 47% carries wide uncertainty the paper does not quantify. Checked the PubMed abstract and the PMC full text (PMC7594389): this correlative paper carries no safety section, so toxicities[] is empty and the parent-trial toxicity sits on the Uy 2021 row. — (toxicity table not extracted)

Felices/Miller (2020) — Cancer Immunol Res

Reference: [PMID 32661096](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Second-generation CD33 TriKE (cam1615CD33, the molecule developed as GTB-3650)
- **Indication / model:** human NK cells with CD33-expressing leukaemia targets in vitro; preclinical mouse tumour-control models
- **Design:** Three arms in one molecule: a humanised single-domain antibody against CD16 to trigger NK activation, an anti-CD33 arm for target recognition, and an IL-15 moiety that drives NK expansion and priming without systemic cytokine exposure. NK-mediated killing does not require HLA presentation, which matters when class I/II escape is on the table.
- **Effect size:** moderate — Replacing the anti-CD16 scFv with a single-domain antibody produced stronger and more NK-selective proliferation without T-cell stimulation, higher killing of CD33-expressing targets, and better tumour control in mouse models. Selectivity for NK over T cells is the practical advance.
- **Variance:** vs first-generation scFv-anti-CD16 TriKE; NK cells alone; IL-15 alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Antigen-density dependence was not systematically dissected, so the pending CD33 quantitation cannot be read against these data. NK function in a patient six years past allograft and through FLAG-IDA is unknown.

Maslah/Giraudier (2020) — Haematologica

Reference: [PMID 31488557](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246/PRIMA-1Met) plus azacitidine in TP53-mutated MDS/AML
- **Indication / model:** TP53-mutated MDS/AML cell lines, TP53-mutated primary patient cells, and in-vivo models; transcriptomic profiling of treated cells
- **Design:** APR-246 converts to methylene quinuclidinone, which alkylates cysteines in the mutant p53 core domain and restores transactivation. Combining with azacitidine adds hypomethylation, and the transcriptomic readout points to downregulation of FLT3 signalling as the shared node.
- **Effect size:** moderate — Low-dose APR-246 reactivated the p53 pathway and induced apoptosis, synergising with azacitidine in TP53-mutated cell lines, primary MDS/AML cells and in vivo. Adding FLT3 ligand reversed the antiproliferative effect, identifying FLT3 pathway suppression as the mediator of synergy.
- **Variance:** vs untreated cells; single-agent APR-246; single-agent azacitidine; FLT3-ligand rescue arm; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Mechanism is broad mutant-p53 refolding rather than variant-selective, which is why eprenetapopt reads as low-yield here — its randomised clinical programme has already failed its primary endpoint. Applies only if TP53 aberration is confirmed.

Pei/Jordan (2020) — Cancer Discov

Reference: [PMID 31974170](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Venetoclax — monocytic-differentiation resistance and MCL1 dependence
- **Indication / model:** paired patient specimens across venetoclax/azacitidine response, single-cell and bulk transcriptomics, primary AML subpopulations sorted by developmental phenotype, metabolic and BH3 profiling
- **Design:** Monocytic AML cells drop BCL2 expression, run oxidative phosphorylation off MCL1 instead, and so survive BCL-2 inhibition; venetoclax then acts as a selection pressure that lets those subclones expand at relapse.
- **Effect size:** strong — Response tracked developmental stage — primitive AML was sensitive and monocytic AML resistant — with resistant cells showing a distinct transcriptome, loss of BCL2, and MCL1-dependent survival. Relapse specimens were enriched for the monocytic subpopulation.
- **Variance:** vs phenotypically primitive AML populations from the same patients; responders versus non-responders; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** This patient's blasts are CD34-positive and CD14/CD64-negative, so the monocytic route probably does not explain her venetoclax failure — the TP53/apoptotic-network route in Nechiporuk 2019 fits better. Logged so the board can rule it out on immunophenotype rather than assume it.

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MDG1021 (HA-1H TCR-modified donor T cells)
- **Indication / model:** Relapsed or persistent haematologic malignancy after allogeneic HSCT (with or without unmanipulated DLI) (2L+); HLA-A02:01 positive; HA-1H positive recipient; sibling or unrelated donor HA-1H-negative or single-mismatched at HLA-A02:01

- **Design:** Dose-escalation phase 1 of donor-derived HA-1H TCR-modified T cells; withdrawn before enrolling
- **n:** 0
- **Effect size:** — — Safety, feasibility, preliminary efficacy
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) +/- pembrolizumab
- **Indication / model:** Locally advanced or metastatic solid tumours harbouring TP53 Y220C (PYNNALE) (2L+); TP53 Y220C mutation; KRAS single-nucleotide variants excluded from phase 2
- **Design:** Open-label multicentre phase 1/2 of rezatapopt as monotherapy and with pembrolizumab
- **n:** 300
- **Effect size:** — — Safety, tolerability, objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Chapuis/Greenberg (2019) — Nature Medicine

Reference: [PMID 31235963](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** WT1-TCRC4 CD8+ T cells (TTCR-C4, HLA-A*02:01-restricted), prophylactic post-HCT
- **Indication / model:** High-risk AML in remission after allogeneic HCT (maintenance); 12 HLA-A2-positive patients transplanted for poor-risk AML and infused while in remission; contemporaneous comparison group of 88 similar-risk AML patients from the same programme.
- **Design:** Phase 1/2 single-arm with contemporaneous non-randomised comparison group (NCT01640301)
- **n:** 12
- **Effect size:** 100 % — Relapse-free survival after infusion
- **Variance:** p=0.002
- **Toxicities:** CRS (0/12, 0%); Pulmonary toxicity (0/12, 0%); Acute GVHD G3 (1/12, 8.3%); Chronic GVHD (n=12, 55%)
- **Case fit:** weak
- **Notes:** Prophylactic infusion into patients in remission. Read against Mazziotta 2025, which applied the same product to active post-HCT disease and did not reproduce the benefit; that is the population this patient sits in.

Pemmaraju/Konopleva (2019) — New England Journal of Medicine

Reference: [PMID 31018069](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tagraxofusp (SL-401, CD123-directed diphtheria-toxin fusion protein)
- **Indication / model:** Blastic plasmacytoid dendritic cell neoplasm, untreated or relapsed (1L); 47 patients assigned across 7 and 12 microg/kg cohorts; the primary-outcome population was the 29 previously untreated patients dosed at 12 microg/kg. Median age 70.
- **Design:** Open-label multicohort phase 2 (NCT02113982)

- **n:** 29
- **Effect size:** 72 % — Complete response or clinical complete response
- **Toxicities:** Capillary leak syndrome (n=47, 19%); ALT increased (n=47, 64%); AST increased (n=47, 60%); Hypoalbuminaemia (n=47, 55%); Peripheral oedema (n=47, 51%); Thrombocytopenia (n=47, 49%)
- **Case fit:** cross_tumor_only
- **Notes:** Registration trial for the only approved CD123-directed agent, in BPDCN rather than AML. It is the reason tagraxofusp exists as an off-label CD123 option and nothing more; the 72% figure does not transfer to AML.

Laszlo/Walter (2019) — Haematologica

Reference:PMID 30115657 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin vs CD33/CD3 BiTE AMG 330 — dependence on CD33 antigen density
- **Indication / model:** pretreatment primary specimens from 45 adults with AML, assayed ex vivo for CD33 surface density, rs12459419 genotype, and cytotoxicity from gemtuzumab ozogamicin and from AMG 330 with autologous T cells
- **Design:** An ADC has to internalise enough conjugate to deliver a lethal calicheamicin dose, so surface CD33 acts as a proxy for intracellular payload. A T-cell engager only needs enough antigen to form a functional immune synapse, which is a much lower bar.
- **n:** n=45 primary AML specimens (rs12459419: CC 31.1%, CT 57.8%, TT 11.1%)
- **Effect size:** moderate — CD33 density correlated with gemtuzumab ozogamicin cytotoxicity ($r^2=0.3124$, $P<0.0001$) but not with AMG 330 killing ($r^2=0.0136$, $P=0.45$ at 100 pg/mL). The rs12459419 genotype shifted CD33 expression roughly ten-fold (CC 2,344 $\pm$ 579 vs TT 255 $\pm$ 91 MFI) yet did not by itself predict cytotoxicity for either agent.
- **Variance:** vs drug-free wells; comparison across the two agents on the same specimens; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** This is the paper that answers how much the pending quantitative CD33 flow actually matters: density is a real but partial determinant for the ADC (about a third of the variance) and close to irrelevant for a T-cell-redirecting format. Ex-vivo cytotoxicity, not patient outcome.

Nechiporuk/Tyner (2019) — Cancer Discov

Reference:PMID 31048320 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Venetoclax / BCL-2 inhibition — TP53-dependent resistance (prior exposure in this case)
- **Indication / model:** genome-wide CRISPR/Cas9 resistance screen in AML cell lines with validation knockouts; cross-referenced against Beat AML patient drug-response and expression data
- **Design:** Venetoclax kills by freeing BIM and BAX from BCL-2, so loss of TP53, BAX or PMAIP1 removes the effector machinery downstream of target engagement. Resistant cells also lower BCL2 and lean on BCL2L1, with accompanying shifts in mitochondrial homeostasis and metabolism.
- **Effect size:** strong — TP53, BAX and PMAIP1 knockouts each conferred venetoclax resistance, and resistance was accompanied by reduced BCL2 expression and reliance on alternative family members. TP53-knockout cells gained sensitivity to TRK inhibitors, an unexpected collateral vulnerability.
- **Variance:** vs parental unedited lines; non-targeting guides; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong

- **Notes:** This is the most direct mechanistic account of why FLAG-IDA plus venetoclax failed in a TP53-aberrant patient, and it argues against re-treating with a venetoclax backbone. Isogenic knockouts model clean biallelic loss rather than the subclonal or copy-loss pattern likely here.

Toffalori/Vago (2019) — Nat Med

Reference: [PMID 30911134](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Post-transplant relapse — CIITA-driven HLA class II loss plus inhibitory ligand upregulation
- **Indication / model:** purified AML blasts from serial timepoints across the disease history, genomic and expression profiling, validated in two independent patient cohorts with paired circulating donor T-cell phenotyping
- **Design:** Two separable escape routes converge on losing donor T-cell recognition: downregulation of the class II master regulator CIITA silences HLA-DR, -DQ and -DP, while PD-L1, B7-H3, CD80 and PVRL2 rise to deliver inhibitory signals. Interferon-gamma reverses the first, checkpoint blockade the second.
- **Effect size:** strong — A transcriptional signature specific to post-transplant relapse was heavily enriched for T-cell costimulation and antigen presentation, with coordinated changes in circulating donor T cells. Loss of class II expression and upregulation of inhibitory checkpoints behaved as alternative, largely non-overlapping escape modalities.
- **Variance:** vs the same patients' pre-transplant and chemotherapy-relapse timepoints; independent validation cohorts; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Overlaps substantially with Christopher 2018 — kept because it adds the CIITA mechanism, the checkpoint-ligand arm, and the pharmacological reversal that make the interferon-gamma plus DLI trial option (NCT06529731) mechanistically coherent for this patient.

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pivkemab sunirine +/- venetoclax +/- azacitidine
- **Indication / model:** CD123-positive AML, relapsed/refractory and treatment-naive cohorts (2L+); CD123 positive by local flow or IHC; prior CD123-directed therapy allowed if CD123 remains detectable
- **Design:** Phase 1b/2 of pivkemab as monotherapy or with venetoclax and/or azacitidine across four regimens
- **n:** 218
- **Effect size:** — — Safety and composite complete remission rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FLAG-GO (cladribine, cytarabine, G-CSF + fractionated gemtuzumab ozogamicin)
- **Indication / model:** Persistent, relapsed or refractory AML including secondary and therapy-related disease (2L+); CD33 expression on at least 20% of blasts by flow cytometry or IHC
- **Design:** Single-arm phase II of cladribine, cytarabine and G-CSF with fractionated gemtuzumab ozogamicin

- **n:** 39
- **Effect size:** — — Complete remission rate
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** 211At-BC8-B10 (astatine-211 anti-CD45 antibody) + haploidentical allo-HCT
- **Indication / model:** High-risk acute leukemia or MDS proceeding to haploidentical allogeneic HCT (2L+); CD45 expression on blasts for patients not in remission; related haploidentical donor mismatched at HLA-A, -B or DRB1 of the unshared haplotype
- **Design:** Dose-escalating astatine-211 anti-CD45 radioimmunotherapy as conditioning ahead of haploidentical transplant with post-transplant cyclophosphamide
- **n:** 30
- **Effect size:** — — Maximum tolerated dose; transplant outcomes
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Ortí/Valcárcel (2018) — Experimental Hematology

Reference: [PMID 29526774](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** DLI versus second allogeneic HCT after debulking therapy
- **Indication / model:** Acute leukemia relapsing after allogeneic HCT (2L+); 46 patients (30 AML, 16 ALL), median age 38; 27 received a second allograft and 19 received DLI, all after debulking.
- **Design:** Retrospective multicentre comparative cohort (Spanish centres)
- **n:** 46
- **Effect size:** no difference on multivariate analysis — — Overall survival, DLI versus second allogeneic HCT
- **Variance:** $p=0.021$
- **Toxicities:** Comparative survival and relapse are the reported outcomes; the paper does not tabulate adverse events by term.
- **Case fit:** partial
- **Notes:** Small (46 patients) and underpowered, so the equivalence claim is weak. Its use is that once debulking is applied to both, remission duration rather than the choice between DLI and a second graft drove outcome. Checked PubMed abstract and the Experimental Hematology page; the retrospective cohort reports survival and relapse only, with no per-term AE table and no registration record.

Dossa/Bleakley (2018) — Blood

Reference: [PMID 29051183](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HA-1 TCR-T (donor-derived CD8+/CD4+ memory T cells, Bleakley/HighPassBio platform)
- **Indication / model:** human primary CD4+ and CD8+ memory T cells lentivirally transduced with the HA-1 TCR transgene; cytotoxicity against primary HLA-A*02:01+ HA-1H+ leukemia blasts

- **Design:** An HLA-A*02:01-restricted TCR against HA-1H, a minor histocompatibility antigen whose expression is confined to haematopoietic tissue, so donor-derived T cells kill recipient leukaemia without the broad tissue toxicity of unselected alloreactivity. The transgene also carries a CD8 coreceptor (to make the class-I TCR work in CD4+ cells), an inducible caspase-9 kill switch, and a CD34-CD20 selection/tracking epitope.
- **Effect size:** moderate — Both CD4+ and CD8+ transduced memory T cells recognised and killed primary HA-1H-positive leukaemia; naive T cells were excluded from the product to limit native-TCR alloreactivity. This is the construct behind NCT03326921, so the preclinical package and the trial product are the same object.
- **Variance:** vs untransduced and irrelevant-TCR-transduced T cells from the same donors; HA-1-negative and HLA-A2-negative targets; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro human work only — no xenograft efficacy arm in this paper. Applicability here is entirely conditional on retrieving HLA-A*02:01 status plus recipient HA-1(H)-positive / sibling-donor HA-1-negative genotyping, neither of which is on hand.

Kovtun/Chittenden (2018) — Blood Adv

Reference:PMID 29661755 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Pivekimab sunirine (IMGN632, CD123-directed antibody-drug conjugate)
- **Indication / model:** AML cell lines and patient-derived AML samples in culture; normal human myeloid progenitors; multiple AML xenograft models
- **Design:** A humanised anti-CD123 antibody delivers a DNA mono-alkylating indolinobenzodiazepine pseudodimer. Swapping the crosslinking payload for a mono-alkylator is what buys the therapeutic window, since crosslinking IGNs are far more toxic to normal progenitors.
- **Effect size:** strong — Both conjugates reduced AML viability at low picomolar concentrations irrespective of multidrug-resistance phenotype or disease status, but the crosslinking comparator was more than 40-fold more toxic to normal myeloid progenitors than IMGN632. Antitumour activity held across several AML xenograft models.
- **Variance:** vs X-ADC, the same G4723A antibody carrying a DNA-crosslinking IGN payload; unconjugated antibody; normal bone marrow progenitors; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Sponsor-run study (ImmunoGen). Relapsed/refractory post-transplant disease and extramedullary sites are not represented in the model set.

Campagne/Mager (2018) — Clin Cancer Res

Reference:PMID 29463552 · **Type:** preclinical · **Inclusion:** excluded — Pharmacokinetic/pharmacodynamic modelling in healthy cynomolgus monkeys with no leukaemia and no antileukaemic endpoint; the human efficacy question is already answered by the clinical rows for this agent.

- **Intervention:** Flotetuzumab (MGD006, CD123 x CD3 DART) — nonhuman-primate PK/PD
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Pharmacokinetic/pharmacodynamic modelling in healthy cynomolgus monkeys with no leukaemia and no antileukaemic endpoint; the human efficacy question is already answered by the clinical rows for this agent.

Christopher/DiPersio (2018) — N Engl J Med

Reference: [PMID 30380364](#) · Type: preclinical · Inclusion: included

- **Intervention:** Post-transplant AML relapse — MHC class II downregulation as immune escape
- **Indication / model:** paired presentation/relapse primary AML samples analysed by exome and RNA sequencing, flow cytometry and immunohistochemistry, with in-vitro interferon-gamma treatment of blasts
- **Design:** Selection pressure from donor T cells favours blasts that stop presenting antigen to CD4+ cells; MHC class II genes are transcriptionally downregulated rather than mutated or deleted, which is why exome sequencing misses it.
- **n:** 15 patients relapsing after HCT (matched-sibling, matched-unrelated or mismatched-unrelated) vs 20 relapsing after chemotherapy; 34 post-HCT relapses in the extended flow/IHC set
- **Effect size:** strong — Post-transplant relapse carried no relapse-specific mutations in immune genes, but HLA-DPA1, HLA-DPB1, HLA-DQB1 and HLA-DRB1 fell 3- to 12-fold, and reduced MHC class II protein was confirmed in 17 of 34 post-HCT relapses. Interferon-gamma reversed the phenotype rapidly in vitro.
- **Variance:** vs relapse after chemotherapy rather than transplant; paired presentation samples from the same patients; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The cohort includes matched-sibling grafts, so unlike genomic HLA-haplotype loss this mechanism is available to this patient. It is a class II phenomenon, so it degrades CD4-mediated graft-versus-leukaemia and DLI more than it degrades a class-I-restricted HLA-A*02:01 TCR product.

—/— (2018)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** MDG1011 (PRAME TCR-modified autologous T cells)
- **Indication / model:** High-risk myeloid and lymphoid neoplasms (AML, MDS, MPN, multiple myeloma) (2L+); HLA-A*02:01 positive by genotyping; PRAME expression
- **Design:** Open-label dose escalation with a concurrent HLA-A*02:01-negative control group; stopped after phase I
- **n:** 9
- **Effect size:** — — Safety, tolerability, feasibility
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Tawara/Shiku (2017) — Blood

Reference: [PMID 28860210](#) · Type: clinical · Inclusion: included

- **Intervention:** WT1-siTCR gene-transduced lymphocytes (TBI-1301, HLA-A*24:02-restricted)
- **Indication / model:** Refractory AML and high-risk MDS (2L+); 8 HLA-A24:02-positive patients with refractory AML or high-risk MDS across two dose cohorts. Restriction is A24:02, so this route stays open if A*02:01 typing returns negative.
- **Design:** First-in-human phase 1, single-arm (UMIN000011519)
- **n:** 8

- **Effect size:** 25 % — Transient reduction in marrow blasts
- **Toxicities:** Normal-tissue adverse event (0/8, 0%)
- **Case fit:** partial
- **Notes:** Activity signal is a transient blast reduction in 2 of 8, not remission. Its value here is the alternative HLA restriction: it is the one WT1 handle that does not require A*02:01. No per-term AE table is published beyond the statement that normal-tissue events did not occur.

Malard/Nagler (2017) — Biology of Blood and Marrow Transplantation

Reference: [PMID 27816650](#) · **Type:** clinical · **Inclusion:** excluded — Wrong disease state — all 265 patients were transplanted in CR1 or CR2, whereas this patient has 85% marrow blasts. The refractory-disease FLAMSA evidence is Schmid 2005.

- **Intervention:** FLAMSA sequential conditioning in AML in first or second complete remission (EBMT)
- **Notes:** Excluded: Wrong disease state — all 265 patients were transplanted in CR1 or CR2, whereas this patient has 85% marrow blasts. The refractory-disease FLAMSA evidence is Schmid 2005. — Read for a larger FLAMSA denominator. The 2-year OS of 56.1% here belongs to remission patients and should not be quoted for active disease.

Arcangeli/Varani (2017) — Mol Ther

Reference: [PMID 28479045](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CD123-directed CAR T cells — affinity and antigen-density thresholds
- **Indication / model:** anti-CD123 CAR T cells with structure-guided affinity mutations in the binding domain, tested against AML lines and CD123-low healthy targets across a range of antigen densities
- **Design:** CAR output is set by the product of binding affinity, CAR surface expression and target antigen density; lowering affinity raises the antigen threshold needed to trigger a response, which is how on-target/off-tumour recognition of CD123-low normal tissue can be dialled down.
- **Effect size:** moderate — Early cytotoxicity was largely insensitive to CAR affinity or expression, but later effector functions such as cytokine release and proliferation failed when CAR expression was low. Distinct lytic and activation antigen thresholds emerged, and lowering affinity preserved anti-AML killing while improving the normal-tissue margin.
- **Variance:** vs parental high-affinity CAR; CAR-negative T cells; CD123-low normal endothelium and progenitor targets; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In vitro threshold work with no leukaemia-burden endpoint, and the constructs are not the Allo-RevCAR01/R-TM123 system on the trial list. It gives the board a framework for reading the pending quantitative CD123 flow, not a prediction for a specific product.

—/— (2017)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GLIDE (guided lymphocyte immunopeptide-derived expansion; donor anti-minor-histocompatibility-antigen T cells)
- **Indication / model:** Molecular or clinical relapse of haematologic malignancy after matched-donor HSCT

(GLIDE) (2L+); Any of a 19-allele HLA panel including HLA-A02:01 and HLA-A24:02; targetable minor histocompatibility antigen identified by paired patient/donor exome sequencing

- **Design:** Exploratory open-label multicentre study of ex vivo expanded donor anti-MiHA T lymphocytes
- **n:** 20
- **Effect size:** — — Safety and efficacy
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2017)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BPX-701 (PRAME TCR-T with iCasp9 switch) + rimiducid
- **Indication / model:** AML relapsed after first CR, previously treated MDS, or metastatic uveal melanoma (2L+); HLA-A2*01 positive by local testing; PRAME expression by central testing
- **Design:** Dose-finding study of a PRAME TCR-T carrying an inducible caspase-9 safety switch activated by rimiducid
- **n:** 4
- **Effect size:** — — Safety, feasibility, activity
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2017)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tagraxofusp (SL-401) + azacitidine +/- venetoclax
- **Indication / model:** Relapsed or refractory AML (cohort B), high-risk MDS (cohort A), BPDCN (cohort C) (2L+); CD123 / IL3RA expression on blasts determined locally within 3 months
- **Design:** Phase 1 of SL-401 with azacitidine or azacitidine plus venetoclax across three disease cohorts
- **n:** 72
- **Effect size:** — — Safety, tolerability, maximum tolerated dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2016)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Donor-derived multiTAA-specific T cells (ADSPAM)
- **Indication / model:** AML or MDS after any allogeneic HSCT, as adjuvant therapy or for relapsed/residual disease (2L+); Multi-antigen specificity (WT1, PRAME, survivin, NY-ESO-1); donor-derived product
- **Design:** Two-group phase 1 (adjuvant vs relapsed/MRD) of donor-derived multi-tumour-antigen-specific T cells
- **n:** 44
- **Effect size:** — — Safety and feasibility

- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Wang/Han (2015) — Molecular Therapy

Reference:PMID 25174587 · **Type:** clinical · **Inclusion:** included

- **Intervention:** CD33-directed CAR T cells (CART-33)
- **Indication / model:** Refractory AML (2L+); Single 41-year-old male patient with refractory AML and pre-existing pancytopenia.
- **Design:** Single-patient report from a phase 1 trial (NCT01864902)
- **n:** 1
- **Effect size:** marked decrease at 2 weeks, progression by 9 weeks — — Marrow blast reduction
- **Toxicities:** Chills and fever (1/1, 100%); Cytokine elevation (IL-6, IL-8, TNF-alpha, IFN-gamma) (1/1, 100%); Hyperbilirubinaemia G1-2 (1/1, 100%); Pancytopenia fluctuation G>=3 (1/1, 100%)
- **Case fit:** weak
- **Notes:** A single patient is the whole published clinical experience for CD33 CAR-T in adults. Anything the board says about NCT05672147 rests on this plus preclinical work.

Ruutu/Kröger (2015) — Bone Marrow Transplantation

Reference:PMID 26367221 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Second allogeneic transplant for relapse — EBMT registry with remission-duration stratification
- **Indication / model:** Relapse of malignant disease after a first allogeneic transplant (2L+); 2632 second allogeneic transplants performed for relapse after a first transplant, across diagnoses. Sibling donor was a favourable factor; changing donor was not.
- **Design:** Retrospective registry analysis (EBMT)
- **n:** 2632
- **Effect size:** 15 % — Relapse-free survival at 5 years after the second transplant
- **Toxicities:** Adverse events are not tabulated in this registry analysis; the outcome measures are survival and relapse.
- **Case fit:** partial
- **Notes:** The largest denominator on second transplants and the source of the stratification this case turns on: a six-year remission and a sibling donor sit on the favourable side of every listed factor except disease burden, which sits badly at 85% blasts. Effect sizes for individual strata are not published as separate survival estimates. Checked PubMed abstract and the Bone Marrow Transplantation page; EBMT registry analyses of this type carry no AE data and there is no registration record.

Fujiwara/Yasukawa (2015) — Leukemia

Reference:PMID 26104661 · **Type:** preclinical · **Inclusion:** excluded — In-vitro characterisation of CD4+ helper function from the same laboratory; the CD4-augmentation argument it makes is already carried, in the product actually on the trial list, by the CD8-coreceptor design in Dossa 2018.

- **Intervention:** WT1-specific TCR-redirection CD4+ T cells (class-I-restricted TCR in CD4 compartment)
- **Toxicities:** n/a (preclinical)

- **Notes:** Excluded: In-vitro characterisation of CD4+ helper function from the same laboratory; the CD4-augmentation argument it makes is already carried, in the product actually on the trial list, by the CD8-coreceptor design in Dossa 2018.

—/— (2015)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Gene-modified WT1 TCR T cells (Cell Medica)
- **Indication / model:** MDS or AML failing to achieve or maintain an IWG response after hypomethylating-agent therapy (2L+); HLA-A*02:01 restricted; WT1 expression
- **Design:** Single-arm phase I/II of autologous WT1 TCR-transduced T cells
- **n:** 3
- **Effect size:** — — Safety and efficacy
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2015)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tumour-associated-antigen-specific T cells (TAA-T)
- **Indication / model:** Very high-risk AML and MDS around allogeneic HSCT (2L+); Multi-antigen specificity (WT1, PRAME, survivin and others); T-cell chimerism >94% if recipient-derived
- **Design:** Multi-institutional phase I of ex vivo expanded tumour-associated-antigen-specific lymphocytes given after transplant
- **n:** 50
- **Effect size:** — — Safety and feasibility
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Hills/Burnett (2014) — Lancet Oncology

Reference: [PMID 25008258](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin added to induction — individual-patient-data meta-analysis
- **Indication / model:** Adult AML (excluding acute promyelocytic leukaemia) (1L); 3325 patients across five randomised trials, with a prespecified exploratory stratification by cytogenetic risk group.
- **Design:** Individual-patient-data meta-analysis of randomised controlled trials
- **n:** 3325
- **Effect size:** 0.99 OR — Six-year absolute survival benefit by cytogenetic risk
- **Variance:** 95% CI 0.83–1.18; p=0.9
- **Toxicities:** Early death G5
- **Case fit:** weak
- **Notes:** Carries the caveat the board most needs on CD33: patients with adverse cytogenetics showed no survival benefit (OR 0.99, 0.83-1.18, p=0.9). This patient's karyotype is complex and adverse.

Orozco/Pagel (2013) — Blood

Reference:PMID 23471305 · **Type:** clinical · **Inclusion:** excluded — Preclinical — disseminated murine AML model with syngeneic transplant; no clinical results are published for NCT03670966.

- **Intervention:** 211At-BC8-B10 (astatine-211 anti-CD45) conditioning
- **Notes:** Excluded: Preclinical — disseminated murine AML model with syngeneic transplant; no clinical results are published for NCT03670966. — Logged so the thinness is visible: the alpha-emitter conditioning protocol that would accept this patient's prior reduced-intensity allograft has no published human efficacy data at all.

Christopeit/Schmid (2013) — Journal of Clinical Oncology

Reference:PMID 23918951 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Second allogeneic HCT for haematologic relapse of acute leukemia
- **Indication / model:** Acute leukemia relapsing after a first allogeneic HCT (2L+); 179 second transplants, 75 after a matched related first transplant and 104 after an unrelated first transplant. Remission duration after the first transplant and disease stage at the second were the two prognostic factors carried into multivariate analysis.
- **Design:** Retrospective multicentre registry cohort (German registry)
- **n:** 179
- **Effect size:** 25 % — Overall survival at 2 years from the second transplant
- **Variance:** +/- 4% standard error
- **Toxicities:** The report is survival-focused and does not tabulate per-term adverse events; failure was driven by second relapse in about half of those achieving CR.
- **Case fit:** strong
- **Notes:** The stratum that fits this patient is the related-donor first transplant group at 37% +/- 6% two-year survival, roughly half again better than the unrelated group. Changing donor for the second graft neither helped nor hurt. Checked PubMed abstract and the JCO page; this is a registry survival analysis with no adverse-event tabulation and no registration record to fall back on.

Orozco/Pagel (2013) — Blood

Reference:PMID 23471305 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** 211At-anti-CD45 radioimmunotherapy (211At-B10-30F11; the murine analogue of 211At-BC8-B10)
- **Indication / model:** disseminated murine AML in leukaemic SJL/J mice with syngeneic haematopoietic cell transplantation; biodistribution plus survival
- **Design:** An alpha emitter conjugated to anti-CD45 puts short-range, high-linear-energy-transfer radiation onto every haematopoietic cell including the leukaemic clone. The short path length confines the dose relative to beta emitters, which is what allows marrow ablation without proportional organ toxicity.
- **Effect size:** strong — Median survival scaled with dose — 123, 101, 61 and 37 days at 24, 20, 12 and 0 microCi. Conjugate localised to marrow and spleen (18% and 79% injected dose per gram at 24 h) with lower kidney and lung uptake, and white cell nadirs stayed above 2.7 K/microL with recovery by four weeks.
- **Variance:** vs 0 microCi (transplant without radioimmunotherapy); dose-titration arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Syngeneic mouse transplant with a murine anti-CD45 antibody, so neither the allogeneic immunology

nor the human antibody is modelled. No clinical efficacy data exist for 211At-BC8-B10, which makes this the entire evidence base for that trial option.

Castaigne/Dombret (2012) — Lancet

Reference: [PMID 22482940](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Fractionated gemtuzumab ozogamicin added to daunorubicin/cytarabine (ALFA-0701)
- **Indication / model:** Untreated de novo AML, age 50-70 (1L); 280 randomised, 139 analysed per arm. De novo AML only; secondary and treatment-related AML were not enrolled.
- **Design:** Randomised open-label phase 3 (EudraCT 2007-002933-36)
- **n:** 280
- **Effect size:** 0.58 HR — Event-free survival at 2 years
- **Variance:** 95% CI 0.43–0.78; $p=0.0003$
- **Toxicities:** Persistent thrombocytopenia $G \geq 3$ (22/139, 16%); Death from toxicity $G5$ ($n=139$)
- **Case fit:** weak
- **Notes:** The randomised basis for fractionated gemtuzumab and the reason CD33 counts as an actionable handle. Untreated de novo AML in a fitter age band, so it establishes the mechanism rather than the expected benefit here.

Schmid/Mohty (2012) — Blood

Reference: [PMID 22167752](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Salvage of AML relapse after reduced-intensity allogeneic HCT — prognostic score
- **Indication / model:** AML relapsing after reduced-intensity conditioning allogeneic HCT (2L+); 263 relapsed patients drawn from 2815 RIC transplants performed in CR between 1999 and 2008. Prognostic groups were built from remission duration after HCT, marrow blast percentage and prior acute GVHD.
- **Design:** Retrospective registry analysis (EBMT Acute Leukaemia Working Party)
- **n:** 263
- **Effect size:** 14 % — Overall survival at 2 years from relapse
- **Toxicities:** Survival, response and prognostic grouping are the reported outcomes; no per-term adverse-event data appear in the paper.
- **Case fit:** strong
- **Notes:** This patient splits the score: remission far beyond 5 months puts her in the favourable band, 85% marrow blasts puts her in the unfavourable one. Long-term survival occurred almost only where cytoreduction achieved CR before DLI or a second graft, which is the case for treating the blast burden first. Checked PubMed abstract and the Blood page; the registry analysis reports survival and prognostic grouping only, with no AE table and no registration record.

—/— (2012)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Lintuzumab-Ac225 (225Ac-lintuzumab)
- **Indication / model:** Untreated AML in older adults, including secondary and antecedent haematologic disorder (1L); More than 25% of blasts CD33-positive; blast count $\geq 20\%$
- **Design:** Phase I/II of an actinium-225-labelled anti-CD33 antibody, with cytarabine in phase 1

- **n:** 40
- **Effect size:** — — Safety, maximum tolerated dose, remission rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Ochi/Yasukawa (2011) — Blood

Reference:PMID 21673345 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** WT1-siTCR gene-transduced lymphocytes (TBI-1301, HLA-A*24:02-restricted)
- **Indication / model:** human T cells transduced with a retroviral WT1-specific TCR vector carrying siRNAs against endogenous TCR chains; autologous primary leukaemia killing plus a mouse xenograft leukaemia model
- **Design:** siRNA silencing of the endogenous TCR alpha and beta chains suppresses mispairing and raises surface expression of the introduced WT1/HLA-A*24:02 TCR, which increases avidity for WT1 peptide on leukaemia cells.
- **Effect size:** moderate — siTCR-transduced T cells outperformed the conventional WT1 TCR construct, lysed autologous patient leukaemia while sparing normal haematopoietic progenitors, and controlled leukaemia in xenografted mice without inhibiting human haematopoiesis. Repeated WT1-peptide stimulation expanded the cells without loss of specificity.
- **Variance:** vs conventional codon-optimised WT1 TCR (WT1-coTCR) transduced T cells; normal CD34+ haematopoietic progenitors as an on-target/off-tumour readout; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Restriction is HLA-A24:02, so this arm only opens if the pending typing returns A24:02 — it is the fallback if A*02:01 is negative, not a parallel option.

Vago/Fleischhauer (2009) — N Engl J Med

Reference:PMID 19641204 · **Type:** preclinical · **Inclusion:** excluded — The escape mechanism it describes — acquired uniparental disomy of 6p deleting the donor-mismatched HLA haplotype — needs a mismatched haplotype to delete. This patient's graft came from an HLA-identical sibling, so there is no mismatched haplotype for the leukaemia to lose, and the finding does not carry over to her late relapse.

- **Intervention:** Genomic HLA haplotype loss as a post-transplant escape route
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: The escape mechanism it describes — acquired uniparental disomy of 6p deleting the donor-mismatched HLA haplotype — needs a mismatched haplotype to delete. This patient's graft came from an HLA-identical sibling, so there is no mismatched haplotype for the leukaemia to lose, and the finding does not carry over to her late relapse.

Taksin/Castaigne (2007) — Leukemia

Reference:PMID 17051246 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Fractionated gemtuzumab ozogamicin monotherapy (MyloFrance-1)
- **Indication / model:** AML in first relapse (2L+); 57 patients in first relapse. Remission correlated with P-glycoprotein and MRP1 activity, which is the drug-efflux axis rather than CD33 density.

- **Design:** Phase 2, single-arm
- **n:** 57
- **Effect size:** 26 % — Complete remission
- **Toxicities:** Neutropenia $<500/\mu\text{L}$ $G \geq 3$ (n=57); Thrombocytopenia $<50,000/\mu\text{L}$ $G \geq 3$ (n=57); Hepatic toxicity $G \geq 3$ (0/57, 0%); Veno-occlusive disease (0/57, 0%)
- **Case fit:** partial
- **Notes:** The origin of fractionated dosing and the reason gemtuzumab can precede a transplant without veno-occlusive disease. First relapse after chemotherapy, not refractory disease after allo-HCT, so the 26% is an upper bound here.

Schmid/Rocha (2007) — Journal of Clinical Oncology

Reference: [PMID 17909197](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Donor lymphocyte infusion for first haematologic relapse of AML after allogeneic HCT
- **Indication / model:** AML in first haematologic relapse after allogeneic HCT (2L+); 399 patients in first relapse after HCT; 171 received DLI and 228 did not. Risk factors examined within the DLI group included marrow blast burden, karyotype, sex and remission status at infusion.
- **Design:** Retrospective registry comparison (EBMT Acute Leukaemia Working Party)
- **n:** 399
- **Effect size:** 21 % — Overall survival at 2 years
- **Variance:** +/- 3% standard deviation; $p=0.04$
- **Toxicities:** The analysis reports survival and risk factors rather than a per-term adverse-event table; GVHD after DLI is discussed narratively as the mechanism of both benefit and harm.
- **Case fit:** strong
- **Notes:** Sets the honest ceiling on sibling DLI here. Late relapse is in this patient's favour, but 85% blasts and an adverse karyotype put her in the 15% +/- 3% stratum, not the 56% one, unless cytoreduction gets her into remission first. Checked PubMed abstract and the JCO page; GVHD is discussed as mechanism rather than tabulated, and the registry study has no registration record to query.

Schmid/Kolb (2005) — Journal of Clinical Oncology

Reference: [PMID 16110027](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** FLAMSA-RIC (sequential fludarabine/cytarabine/amsacrine, reduced-intensity conditioning, prophylactic DLI)
- **Indication / model:** High-risk AML and MDS, including progressive or refractory disease (2L+); 75 consecutive patients, median age 52.3. 59 had progressive or refractory disease; unfavourable karyotype in 49% of informative patients; 68 had contraindications to standard conditioning. 31 had HLA-identical sibling donors.
- **Design:** Prospective single-arm cohort
- **n:** 75
- **Effect size:** 88 % — Complete remission after sequential conditioning
- **Toxicities:** The report gives no per-term adverse-event table; outcome was driven by non-relapse mortality and GVHD, with survival best in patients receiving high CD34+ cell doses and having limited GVHD.
- **Case fit:** partial
- **Notes:** Patients with refractory disease and complex cytogenetics did as well as better-prognosis subgroups here, which is the argument for cytoreduce-then-graft without waiting for remission. Cohort was overwhelmingly first transplant, so it understates the risk of a second graft. Checked PubMed abstract and the JCO publisher

page; the 2005 paper reports non-relapse mortality and GVHD narratively with no per-term AE table, and the study predates trial registration so there is no ClinicalTrials.gov AE tab.

Liu/Frankel (2004) — Exp Hematol

Reference:PMID 15003313 · **Type:** preclinical · **Inclusion:** excluded — Characterises an affinity-enhanced IL-3 variant that was not the molecule taken forward as tagraxofusp, and the endpoints are receptor binding and in-vitro cytotoxicity rather than leukaemia control.

- **Intervention:** DT388IL3 variant with enhanced IL-3 receptor binding
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Characterises an affinity-enhanced IL-3 variant that was not the molecule taken forward as tagraxofusp, and the endpoints are receptor binding and in-vitro cytotoxicity rather than leukaemia control.

Heemskerk/Falkenburg (2003) — Blood

Reference:PMID 12869497 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HA-2-directed TCR gene transfer (mechanism basis for the HA-2 arm, e.g. TSC-101)
- **Indication / model:** peripheral T lymphocytes from HLA-A2-positive HA-2-negative and from HLA-A2-negative donors, retrovirally transduced with HA-2-specific TCR alpha/beta pairs; cytotoxicity against HA-2-expressing leukaemia targets
- **Design:** Transfer of a TCR against HA-2, a second haematopoiesis-restricted minor histocompatibility antigen presented by HLA-A2, converts unselected donor T cells into leukaemia-reactive effectors.
- **Effect size:** moderate — Thirteen HA-2-specific clones shared a conserved Jalpha42 segment, and TCRs transferred into HA-2-negative donor T cells produced antileukaemic lysis. Transfer into HLA-A2-negative donor T cells preserved antileukaemic activity without anti-HLA-A2 alloreactivity.
- **Variance:** vs mock-transduced T cells; HA-2-negative targets; chimeric TCR variants dissecting the Jalpha42 and CDR3 contributions; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Twenty-three-year-old retroviral work, in vitro only, and the TCR is not the one in TSC-101. The HA-2 arm of the current clinical programme rests on a thinner and older preclinical base than the HA-1 arm.

Black/Frankel (2003) — Leukemia

Reference:PMID 12529673 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DT388IL3 (diphtheria toxin-IL-3 fusion; the molecule marketed as tagraxofusp)
- **Indication / model:** irradiated SCID mice inoculated intravenously, intraperitoneally or subcutaneously with v-SRC-transfected TF1 human AML cells
- **Design:** IL-3 fused to the catalytic and translocation domains of diphtheria toxin binds CD123 (IL-3 receptor alpha), internalises, and shuts down protein synthesis by ADP-ribosylating elongation factor 2 — a cell-kill mechanism that does not route through the apoptotic machinery venetoclax and calicheamicin depend on.
- **n:** n=45 (i.v.), n=20 (i.p.), n=20 (s.c.) control animals; 49 DT388IL3-treated i.v. animals
- **Effect size:** strong — Only 5 of 49 intravenously inoculated treated mice died of leukaemia and median disease-free survival exceeded 120 days versus 37 days for controls. The subcutaneous arm behaved very differently: median disease-free survival rose only from 21 to 31 days, so the same dose that near-eradicated

disseminated marrow disease barely dented a solid tissue deposit.

- **Variance:** vs saline; DT388GMCSF; DAB389IL2; cytarabine 80 microg; anti-CD33-calicheamicin conjugate 5 microg; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The subcutaneous-versus-intravenous gap is the closest preclinical analogue on hand for this patient's thigh myeloid sarcoma, and it argues against relying on CD123 immunotoxin alone to clear an extramedullary site. A single transfected cell line in SCID mice, published in 2003.